DO NOW — RECALL

OOP debugging with IDEs

Name: _____ Class: _____ Date: _____

1. Which structure supports BFS traversal?

2. What is an object's field?

3. When a local variable shadows a field, which name is used without qualification?

4. Why might Step Into be more useful than Step Over for a faulty function?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

A FIFO queue.

2. Model answer

Stored state belonging to that object.

3. Model answer

The local variable.

4. Model answer

It lets you inspect statements, parameters and field updates inside the called function.

Vocabulary — know these terms

breakpoint — A chosen instruction where execution pauses.

Step Into — Enter a called function for debugging.

Step Over — Execute a call without entering it line by line.

watch — IDE display of chosen variables/expressions.

object state — Current values of an object's fields and properties.

NEW LEARNING

Read, annotate and explain

Set a breakpoint

Pause at a method call or the following instruction to inspect current state.

Step into to inspect internals

Step Into executes the called method one instruction at a time. Watch Count and parameter n.

Step over to skip internals

Step Over runs the entire call and pauses at the next statement, allowing you to inspect the result.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

A Score class has Points=3. Add(5) should change Points to 8. Identify a wrong line.

Predict first

Step Into Add and inspect the local points and the field Points.

Model the reasoning

The local becomes 8 while the object field remains 3; the assignment targets the wrong variable.

Model the reasoning

Replace it with Points += amount; and rerun. The field should be 8.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Explain Step Into and Step Over at a method call.

Question 2

A class method Increase(int n) contains `int Total = Total + n;` while Total is an existing property. Identify the issue and write correct C# code.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Design an IDE-debugging workflow for a temperature sensor whose Readings list gains two entries after a single AddReading() call.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Step Over a method call means it is not executed.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.